Index

Sl. No	Content	Pg No
01.	Introduction	1
02.	Tools required	1
03.	System design & implementation	2
04.	Implementation & results	3
05.	Results & conclusion	12
06.	Reference	13

Hospital Management using Cisco Packet Tracer

1. INTRODUCTION

In today's digital world, the healthcare industry has significantly transformed through the integration of technology and networking systems. Modern hospitals increasingly rely on digital platforms to manage patient information and support medical activities. Instead of depending on manual records and verbal communication, healthcare institutions utilize secure computer networks to ensure that information flows quickly and accurately between different sections of the hospital.

Networking establishes an efficient communication system among devices and departments such as reception, consulting rooms, laboratories, and administrative units. It enables medical staff to access required information instantly and securely, helping in faster decision-making and improved coordination. By reducing paperwork and human error, the quality of treatment and hospital management greatly improves.

The project focuses on developing a small-scale hospital network system that highlights the role of servers, routers, switches, and workstations in ensuring secure and reliable connectivity. It emphasizes the importance of networking as a backbone of the healthcare system, supporting smooth operations and enhancing medical service delivery. This digital infrastructure demonstrates how technology continues to revolutionize healthcare by making services more efficient, organized, and patient-centered.

2. TOOLS REQUIRED

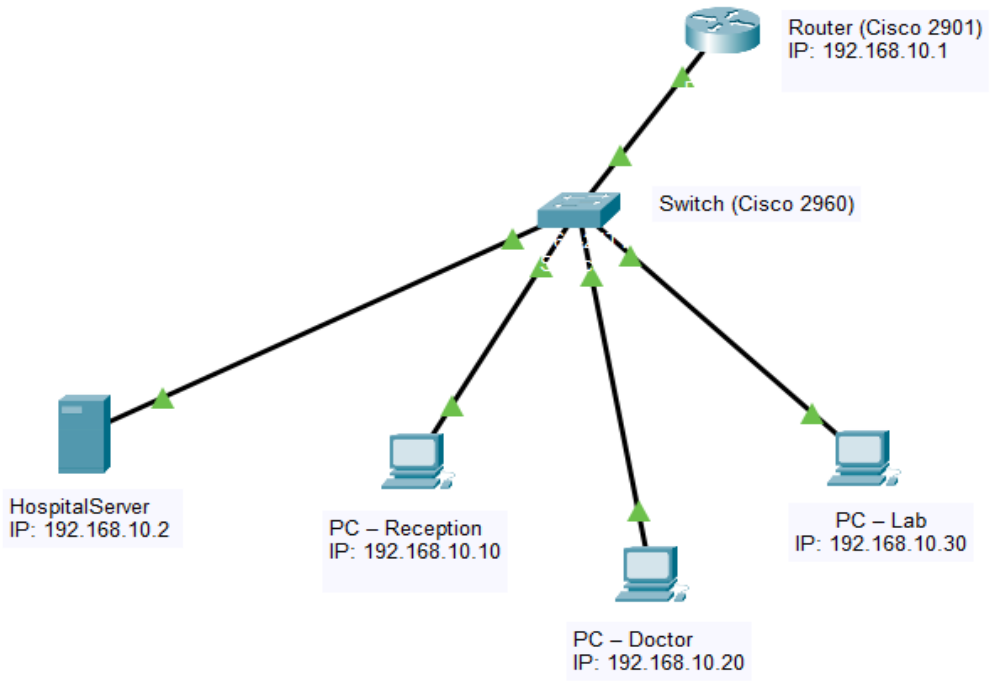
To design and simulate the Hospital Management Network, specific networking tools and devices are used in the Cisco Packet Tracer environment. These resources help build a fully functional virtual LAN environment similar to a real hospital network setup.

Cisco Packet Tracer: Cisco Packet Tracer is a powerful network simulation software developed by Cisco Systems. It provides a virtual environment where users can design, configure, and test computer networks without the need for physical hardware. With a drag-and-drop interface, it allows the placement of routers, switches, servers, PCs, and various networking devices, making it an excellent platform for learning and practicing networking concepts. It supports real-time and simulation modes, helping users visualize how data is transmitted across networks and how devices communicate with each other.

Packet Tracer is widely used in academic environments, especially by students pursuing networking courses such as CCNA. It enables experimentation with routing protocols, IP addressing, services like DNS, DHCP, HTTP, and Email, and various network security features. By providing a practical and interactive learning experience, Cisco Packet Tracer helps learners understand real-world networking scenarios, troubleshoot problems, and build confidence in network design and implementation without requiring expensive equipment.

3. SYSTEM DESIGN & IMPLEMENTATION

3.1 Network Topology Design



3.2 Description of Networking Devices

a) Router (Cisco 2901)
A router is a networking device that forwards data packets between different networks. It manages IP addressing and ensures data reaches the correct destination. In this project, the Cisco 2901 router connects all departments under a unified secure network and acts as the default gateway for communication.

b) Switch (Cisco 2960)
A switch interconnects multiple devices within the same Local Area Network (LAN). It receives data from one device and forwards it only to the intended target device. The Cisco 2960 switch in this design links the Server, Reception PC, Doctor PC, and Lab PC, enabling seamless high-speed internal communication.

c) Hospital Server
A server is a centralized system that stores information and provides services to network clients. In this hospital network, the server hosts essential services like Email and HTTP, which allow departments to access patient records, upload reports, and share prescriptions securely. It ensures reliable data storage and easy access for authorized staff.

d) Personal Computers (PCs)

PCs act as end-user devices operated by hospital staff from different departments:

- **Reception PC:** Used for patient registration and communication with the medical staff.
- **Doctor PC:** Used for checking patient history, receiving lab results, and issuing prescriptions.
- **Lab PC:** Used to upload lab test reports and notify doctors instantly.

4. IMPLEMENTATION & RESULTS

4.1 Device Placement and Naming

- Open Cisco Packet Tracer.
- Place the required network devices from the workspace:
  - 1 Router (Cisco 2901)
  - 1 Switch (Cisco 2960)
  - 1 Server (Generic Server)
  - 3 PCs
- Rename PCs:
  - PC-Reception
  - PC-Doctor
  - PC-Lab

4.2 Adding Connections

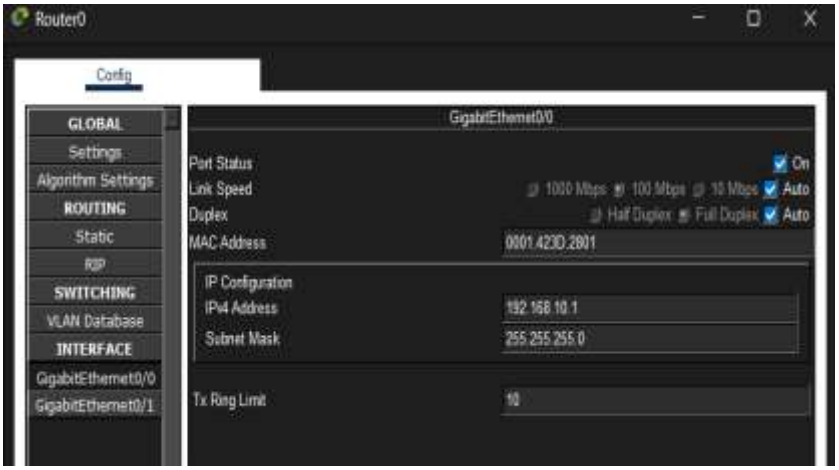
All connections were made using Copper Straight-Through Cable:

From Device	Port	To Device	Port
Router	GigabitEthernet0/0	Switch	GigabitEthernet0/1
Switch	FastEthernet0/2	PC-Reception	FastEthernet0
Switch	FastEthernet0/3	PC-Doctor	FastEthernet0
Switch	FastEthernet0/4	PC-Lab	FastEthernet0
Switch	FastEthernet0/5	Server	FastEthernet0



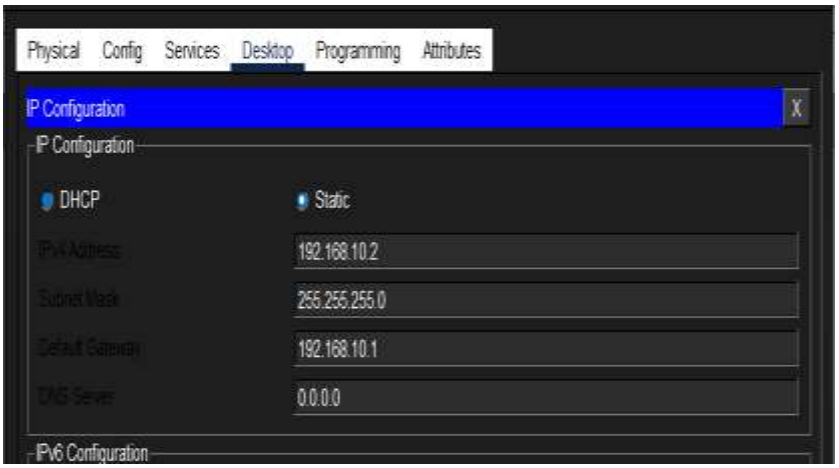
4.3 Router Configuration

- Click Router → Config
- Select **GigabitEthernet0/0**
- Enter:
  - IP Address: **192.168.10.1**
  - SubnetMask: **255.255.255.0**

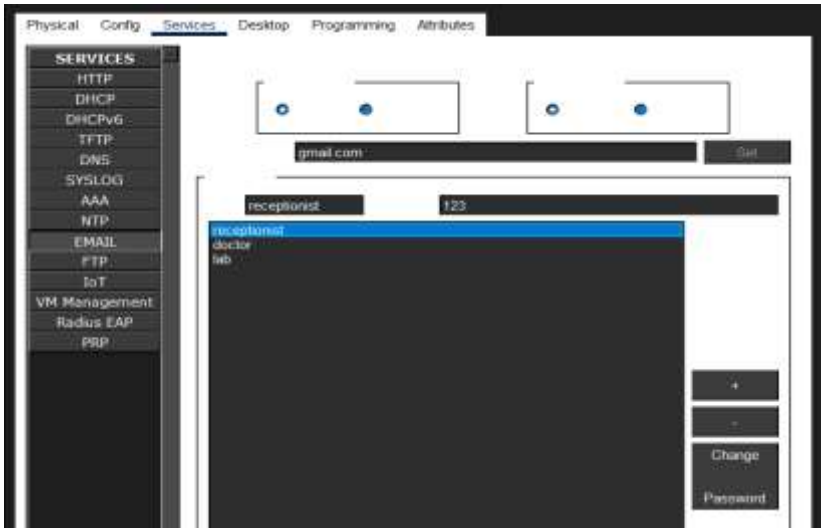


4.4 Server Configuration

- Click Server → Desktop → IP Configuration
  - IP Address: **192.168.10.2**
  - Subnet Mask: **255.255.255.0**
  - Default Gateway: **192.168.10.1**



- Services → Enable:
  - **HTTP**
  - **SMTP**
  - **POP3**
- Services → Email → Create



- User Accounts:

Username	Password	Purpose
receptionist	123	Reception staff
doctor	123	Doctor access
lab	123	Lab technician

4.5 PC IP Configuration

- Click PC → Desktop → IP Configuration

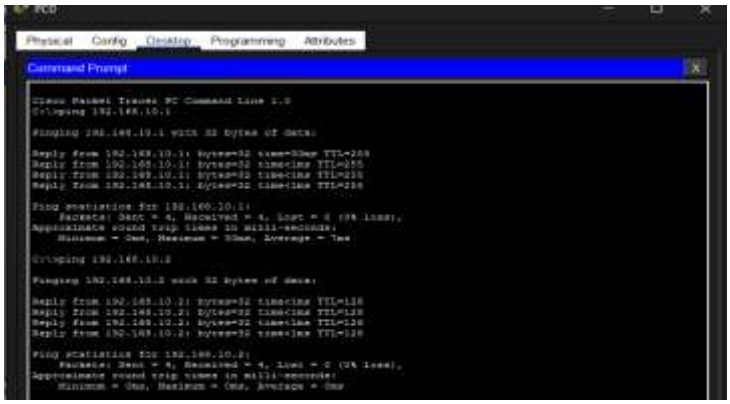
PC Name	IP Address	Gateway	DNS
PC-Reception	192.168.10.10	192.168.10.1	192.168.10.2
PC-Doctor	192.168.10.20	192.168.10.1	192.168.10.2
PC-Lab	192.168.10.30	192.168.10.1	192.168.10.2

4.6 Connectivity Test (Ping Verification)

From each PC:
Open **Command Prompt**
Type: ping 192.168.10.1
ping 192.168.10.2

If you get replies → Everything is correct.





4.7 Email Configuration on PCs

Steps (same for each PC):

- Desktop → **Email** → Configure mail →Add
- Enter:

Field	Reception	Doctor	Lab
Name	Reception	Doctor	Lab
Email	receptionist@gmail.com	doctor@gmail.com	lab@gmail.com
Username	reception	doctor	lab
Password	123	123	123
Incoming Mail (POP3)	192.168.10.2	192.168.10.2	192.168.10.2
Outgoing Mail (SMTP)	192.168.10.2	192.168.10.2	192.168.10.2

4.8 Web Portal Setup on Server

- Open Server → Services → HTTP
- Edit **index.html**

```
<!DOCTYPE html>
<html>
<head>
<title>City Central Hospital Portal</title>

<!-- Styling Section -->
<style>
body {
font-family: Arial;
background-color:#e8f5ff;
padding:20px;
}
h1 {
color:#003d80;
text-align:center;
}
.hospital-img {
text-align:center;
margin-bottom:20px;
}
.record-card {
background:white;
padding:15px;
border-radius:10px;
box-shadow:0px 0px 10px rgba(0,0,0,0.1);
width:60%;
margin:auto;
margin-top:20px;
}
.record-card h2 {
color:#0066cc;
border-bottom:2px solid #cce0ff;
padding-bottom:5px;
}
.details {
font-size:14px;
margin-bottom:10px;
line-height:1.5;
}
a {
color:#0047b3;
text-decoration:none;
font-weight:bold;
}
a:hover {
```

```
color:#0099ff;
text-decoration:underline;
}
</style>

</head>

<body>

<!-- Hospital Image -->
<div class="hospital-img">
<img src="central-hospital-1.jpg" alt="Hospital Building" width="320">
</div>

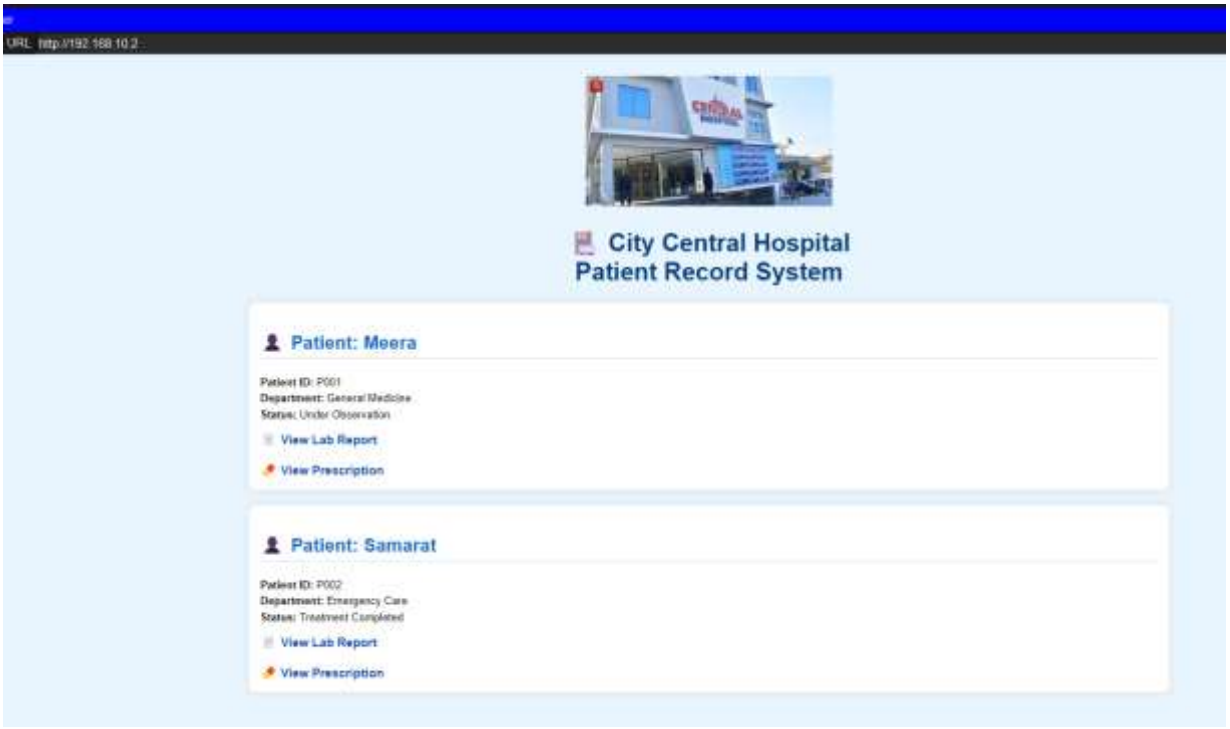
<h1> City Central Hospital<br>Patient Record System</h1>

<!-- Patient 1 -->
<div class="record-card">
<h2> Patient: Meera</h2>
<div class="details">
<b>Patient ID:</b> P001<br>
<b>Department:</b> General Medicine<br>
<b>Status:</b> Under Observation
</div>
<a href="Meera_report.png">  View Lab Report</a><br><br>
<a href="Meera_prescript.png">  View Prescription</a>
</div>

<!-- Patient 2 -->
<div class="record-card">
<h2> Patient: Samarat</h2>
<div class="details">
<b>Patient ID:</b> P002<br>
<b>Department:</b> Emergency Care<br>
<b>Status:</b> Treatment Completed
</div>
<a href="Samarat_Report1.txt">  View Lab Report</a><br><br>
<a href="Samarat_Prescription.txt">  View Prescription</a>
</div>

</body>
</html>
```

- Save
  To access: http://192.168.10.2 on web browser tab.



4.9 Testing Email Workflow

Scenario: Patient Meera needs lab test Meera arrives at the hospital

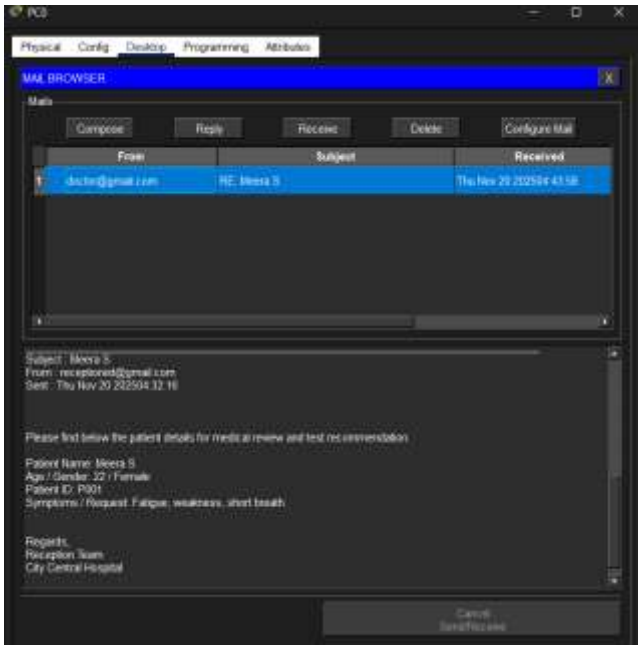
Step A → Reception sends patient details to Doctor

- On Reception PC → Email → **Compose**
- To: doctor@gmail.com

Subject: Patient Details for Review – Meera S (22/F)

Please find below the patient details for medical review and test recommendation.
Patient Name: Meera S
Age / Gender: 22 / Female
Patient ID: P001
Symptoms / Request: Fatigue, weakness, short breath

Regards,
Reception Team
City Central Hospital



Click Send

Doctor PC → Receive email from reception.

Step B → Doctor requests lab report

- Doctor PC → Compose email:
- To: lab@gmail.com

Subject: Test Request – CBC Test for Patient Meera S

Please perform the following diagnostic test for the patient listed below:
Patient Name: Meera S
Patient ID: P001
Test Required: Complete Blood Count (CBC)
Kindly update once the report is ready.

Regards,
Dr. Raghav A (MBBS, MD)
City Central Hospital

Click Send
Lab PC → Receive email from doctor

Step C → Lab uploads report & send mail to Doctor

Click Server
Go to Services → HTTP → import
Meera_report.png from browser and save.

- On Lab PC → Email→ Compose
- To : doctor@gmail.com

Subject: CBC Lab Report Ready – Meera S (P001)

The CBC test report for patient Meera S (Patient ID: P001) has been completed.
The digital report is now available in the hospital server under the patient folder.
Filename: Meera_report.png
You can also view it directly by opening the file in your web browser.
Please review and provide prescription.

Regards,
Sahana M
Lab Technician
City Central Hospital

Click Send
Doctor PC → Receive email from lab

Step D → Doctor uploads prescription & emails Reception
Click Server
Go to Services → HTTP → import
Meera_prescript.png into Server (same as report)

- On Doctor PC → Email→ Compose
- To : receptionist@gmail.com

Subject: Patient Report & Prescription Ready – Meera S (P001)

The lab report for patient Meera (P001) has been reviewed. The prescription has been written and stored on the hospital server.
Files available in the Meera folder:
1. Meera_report.png
2. Meera_Prescript.png
Kindly deliver both files to the patient.

Regards,
Dr. Raghav A (MBBS, MD)
City Central Hospital

Click Send
Reception PC → Receive email from doctor

4.10 Final Output — Report & Prescription Delivery to Patient
After receiving the final confirmation email from the Doctor, the Receptionist proceeds to access the patient records from the Hospital Web Portal.

On PC-Reception:

- Desktop → Web Browser
- Type the Server IP address in the URL bar: http://192.168.10.2

Once verified:

- Physically print the documents
- Hand over both Report and Prescription to patient.

CITY CENTRAL HOSPITAL LABORATORY
CCH Diagnostic Center | NABL Certified | Bengaluru – 560001

Email: lab@gmail.com

LABORATORY TEST REPORT

Patient Name : Meera S

Patient ID : P001

Age / Gender : 22 / Female

Ref. Doctor : Dr. Raghav A (MBBS, MD)

Sample Collected : 22-Nov-2025

Report Date : 22-Nov-2025

Test Type : Complete Blood Count (CBC)

TEST RESULTS

TEST NAME	RESULT	REFERENCE RANGE	FLAG
Hemoglobin (Hb)	10.8 g/dL	12 – 15 g/dL	Low
Hematocrit (HCT)	33.5 %	36 – 46 %	Low
RBC Count	3.94 mil/µL	4.1 – 5.1 mil/µL	Borderline
WBC Count	7,200 /µL	4,000 – 11,000 /µL	Normal
Platelets	2.65 lakh/µL	1.5 – 4.5 lakh/µL	Normal
MCV	82 fl	80 – 100 fl	Normal
MCH	25 pg	27 – 32 pg	Low
MCHC	30.1 g/dL	31.5 – 34.5 g/dL	Slightly Low
RDW	13.2 %	11 – 15 %	Normal

INTERPRETATION

- Mild iron-deficiency anemia indicated due to reduced hemoglobin and MCH.
- Other blood values normal; no sign of infection or platelet abnormality.

SUGGESTIONS

- Iron supplementation recommended
- Iron-rich diet advised (spinach, dates, legumes, beetroot, eggs)
- Repeat CBC after 4 weeks

Authorized By:

Dr. Kavitha R (MD – Pathology)

Lab Technician: Sahana M

NOTE: This is a digital lab report generated electronically. No signature required if sent online.

© 2025 City Central Hospital – Diagnostic Laboratory Information System

CITY CENTRAL HOSPITAL
#45 MG Road, Bengaluru – 560001, Karnataka, India
Email: citycentralhosp@gmail.com

DOCTOR PRESCRIPTION

Patient Name : Meera S
Patient ID : P001
Age / Gender : 22 / Female
Date : 22-Nov-2025
Consultant : Dr. Raghav A (MBBS, MD – General Medicine)

CLINICAL OBSERVATION / DIAGNOSIS
- Complaints of fatigue and physical weakness
- Low MCH & borderline hemoglobin levels in CBC report
- No serious abnormalities detected

PRESCRIPTION (MEDICATIONS)
1) **Ferrous Ascorbate 100 mg**
   - Dosage: Take 1 tablet after breakfast
   - Duration: 30 days

2) **Vitamin B-Complex**
   - Dosage: One tablet after dinner
   - Duration: 30 days

3) **ORS Hydration Pack**
   - Use when tired or dehydrated

DIET & LIFESTYLE ADVICE
- Drink 3–3.5 liters of water every day
- Add spinach, beetroot, dates, dry fruits, eggs, legumes to diet
- Avoid junk food & caffeine
- 30-minute walking / light exercise daily
- Maintain proper sleep cycle – 7 to 8 hours

FOLLOW-UP
- Repeat CBC after 30 days and visit for review

Doctor Signature
Dr. Raghav A
MBBS, MD (General Medicine)

© 2025 City Central Hospital | Confidential Medical Document

5. RESULTS & CONCLUSION

Result:

The Hospital Management Network was successfully designed and implemented using Cisco Packet Tracer. All departments — Reception, Doctor, and Lab — were connected under a secure and unified LAN infrastructure.

The following outcomes were achieved:

✓ All devices (Router, Switch, Server, PCs) were properly configured with correct IP addressing
✓ Successful internal communication within the hospital network
✓ Email system enabled smooth digital communication between departments
✓ Server hosted hospital web portal for secure patient data access
✓ Lab reports and prescriptions were uploaded and accessed through HTTP service
✓ Receptionist was able to view, download, and print patient records to hand over to the patient
✓ System reduced paperwork and processing time significantly.

Conclusion:

This project successfully demonstrates the design and implementation of a secure and efficient hospital communication network using Cisco Packet Tracer. By establishing connectivity between the Reception, Doctor, and Lab departments through centralized server services such as HTTP, SMTP, and POP3, the system ensures reliable exchange of patient information, lab reports, and prescriptions. The digital workflow eliminates manual delays, enhances accuracy, and enables faster decision-making in hospital operations. Overall, the project highlights the importance of networking in healthcare management and provides a practical foundation for expanding into advanced features such as remote access, cloud-based storage, and IoT-enabled healthcare solutions in the future.

****

REFERENCE

1. A.S Tanenbaum - Computer Networks, 4th Edition, PHI, 2003

2. Computer Networks, James J Kurose, Keith W Ross, Pearson Education, 2013, ISBN: 0-273-76896-4

3. Introduction to Data Communication and Networking, Wayarles Tomasi, Pearson Education, 2007, ISBN:0130138282

4. Essentials of Hospital Management & Administration
   Author: Dhoot & Dhoot.